

Ultrastructural and Automated Vocal Fold Tracking Study in Recurrent Laryngeal Nerve Injury Models

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SUMMARY: Objective. This study aims to assess nerve regeneration characteristics and their impact on vocal fold (VF) movement after unilateral recurrent laryngeal nerve (URLN) injuries to inform clinical treatment strategies.

Methods. Four types of URLN injury models in rabbits were created: clamp, electrocautery, transection, and ligation. Nerve damage and regeneration were evaluated using electron microscopy at 2, 4, 8, and 12 weeks postinjury. VF movement range and speed were assessed using tracking software.

Results. The N-ratio (the ratio of myelinated axon area to total nerve trunk area) and myelinated axon density (MAD, the number of axons per unit area): clamp group were the highest at all times compared to the other groups. Postinjury, the electrocautery group was higher than the transection and ligation groups, except at 2 weeks. In the clamp and electrocautery groups, N-ratio and MAD increased between 4 and 8 weeks postinjury. At 4 weeks, 100% of the clamp group and 66.7% of the electrocautery group showed partial recovery of injured VF movement. Compared to the normal VF, the affected VF's mean motion range (MMR) was 22% less ($P < 0.05$), and mean motion speed (MMS) was 58% slower ($P < 0.05$). At 8 weeks, MMR was 7% less ($P < 0.05$), and MMS was 41% slower ($P < 0.05$). At 12 weeks, MAD started to decrease. However, the N-ratio continued to increase. MMR was 2% lower ($P > 0.05$), and MMS was 24% slower ($P < 0.05$).

Conclusion. The severity of RLN injuries was as follows: ligation > transection > electrocautery > clamp injury. Recovery of VF motion range preceded recovery of motion speed. Asynchronous motion of bilateral VFs was a characteristic feature of RLN injury repair.

Keywords: Recurrent laryngeal nerve injury– Transmission electron microscopy– Nerve regeneration– Myelinated axon density– Vocal fold movement.

INTRODUCTION

Recurrent laryngeal nerve (RLN) injuries frequently cause dysphonia, dysphagia, and dyspnea, significantly affecting patients' quality of life. Iatrogenic injuries are a prevalent cause of RLN damage,^{1–3} including clamp, electrocautery, transection, ligation, and traction injuries. During thyroid surgery with neural monitoring, we observed that diverse surgical injuries causing signal attenuation or loss can promptly lead to vocal fold (VF) fixation. However, long-term VF recovery varies: some patients recover fully, while others do not. The type of nerve damage may be the primary factor influencing these differences.

Systematic studies on RLN injuries and associated complications in human patients are infeasible, and the mechanisms remain unclear. Therefore, we utilized a standardized rabbit model of different types of RLN injuries to study the pathogenesis. The objective was to

explore correlations between specific injury mechanisms and VF functional recovery, thereby informing clinical strategies for various types of neural injuries.

Previous methods for evaluating VF movement have included both subjective and objective assessments. Subjective evaluations^{4–8} often fail to detect subtle differences. Objective quantitative methods^{9–12} are limited by still-frame image assessments and do not evaluate VF movement speed. Some researchers^{13,14} have shown that VF tracking software for the automatic quantitative assessment of VF movement is feasible. We developed an automated video tracking system to quantify VF movement speed and amplitude. Using ratio parameters (affected-to-healthy side) for both movement speed and range, we effectively eliminate inter-individual variability and balance confounding factors, enabling reliable longitudinal comparisons.

MATERIALS AND METHODS

Establishment of unilateral recurrent laryngeal nerve (URLN) injury models

Fifty male New Zealand white rabbits (3 months old, $\approx 3 \pm 0.2$ kg) were selected for this study, provided by Hangzhou Fuyang Hongfeng Rabbit Breeding Farm (license number: SCXK (Zhe) 2020-0004). Forty-eight rabbits were used to establish 4 URLN injury model groups (12 rabbits per group), while 2 rabbits served as controls.

Rabbits were anesthetized using a combination of intramuscular injection of Su-Mian-Xin (0.1 mL/kg) and slow intravenous injection of 3% pentobarbital (0.3–0.5 mL/kg) via

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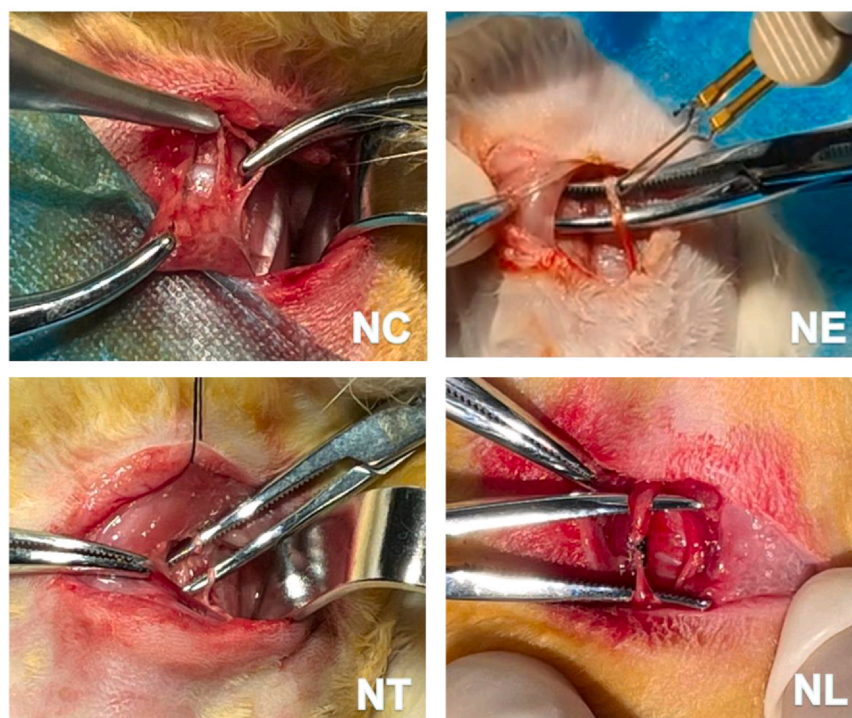


FIGURE 1. Four different types of URLN injury models. NC, nerve clamp injury model; NE, nerve electrocoagulation injury model; NT, nerve transection model; NL, nerve ligation injury model.

the marginal ear vein. Preoperative laryngoscopy was performed to confirm normal bilateral VF movement. At the level of the fourth tracheal ring, the nerve was easily identified and dissected over a length of 1–1.5 cm.

The RLN was injured using different methods according to the experimental group (Figure 1), as follows:

①Nerve clamp injury model (NC): the right RLN was clamped with a hemostatic forceps for one ratchet click, maintained for 1 min, and then released.

②Nerve electrocoagulation injury model (NE): the right RLN was directly cauterized using monopolar coagulation at 1 W power for 1 s, repeated 3 times.

③Nerve transection injury model (NT): the right RLN was completely transected with a scalpel, with the ends left adjacent.

④Nerve ligation injury model (NL): the right RLN was ligated with 4–0 nonabsorbable silk suture.

Immediately after nerve injury, spontaneous VF movement was observed via transoral laryngoscopy to confirm fixation of the injured-side VF and successful model establishment. A black silk suture was placed 1 mm adjacent to the injury site to mark the location of the lesion. This study was approved by the Ethics Committee of Zhongshan Hospital of Xiamen University (ethics number: xmszyky Ethics Review No. 2022-125).

Evaluation of VF motion using automatic tracking software

At 2, 4, 8, and 12 weeks postoperation, three rabbits per group were selected at each time point. Following anesthesia, bilateral VF motion was recorded for 1 minute (24 frames per second)

via transoral laryngoscopy during quiet breathing. A stable and clear 10-second segment from each video was selected for analysis. The video segments were loaded using the built-in MATLAB VideoReader function provided in the core MATLAB R2022a,¹⁵ and frames were accessed sequentially using the readFrame method. Image processing identified key landmarks (anterior commissure and bilateral vocal processes, Figure 2A) and tracked their motion trajectories.

To initialize the tracking procedure, three anatomically defined landmarks—the anterior commissure and the bilateral vocal processes—were identified and manually localized in the first video frame, providing initial coordinate references. Based on these initial landmark coordinates, automatic tracking of the corresponding points in subsequent frames was performed using the PointTracker function from the Computer Vision Toolbox in MATLAB. This procedure yielded frame-by-frame two-dimensional pixel coordinates for each landmark over the selected video segment. Because the anterior commissure remains stationary during VF vibration, it was employed as a reference point for coordinate transformation. Accordingly, the relative motion trajectories of the bilateral vocal processes were computed by subtracting the anterior commissure coordinates from the corresponding absolute coordinates on a frame-by-frame basis, thereby mitigating global translational effects introduced during laryngoscope recording. The relative coordinate data of the tracked landmarks were recorded and exported in CSV format.

The relative coordinate trajectories were subsequently imported into OriginPro (OriginPro 2021)¹⁶ for visualization and post-processing. Trajectory visualization was

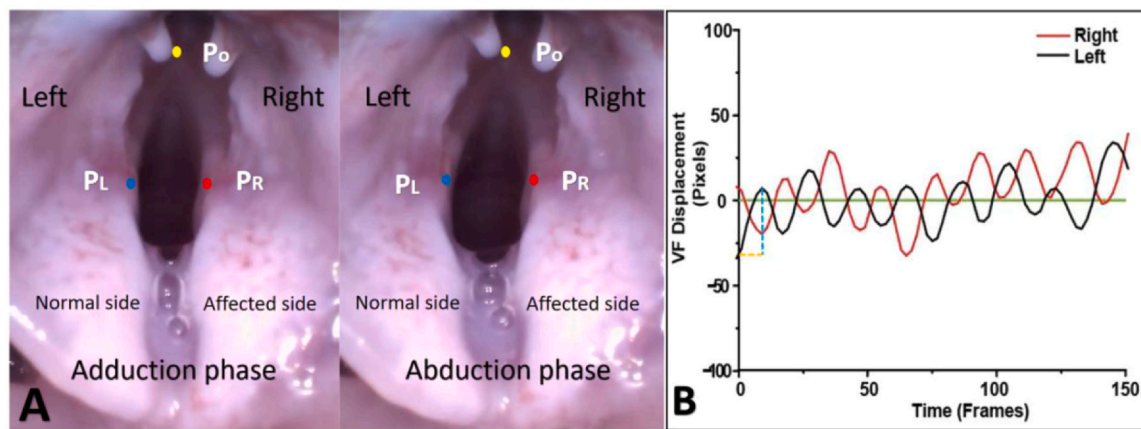


FIGURE 2. Schematic representation of the key point identification methodology and trajectory diagram illustrating normal bilateral vocal fold movement processes. (A) The method for identifying key points (PO, PL, and PR). In this figure, the blue dots mark the normal vocal process (PL), the red dots indicate the affected vocal process (PR), and the yellow dots denote the anterior commissure (PO). The left panel of (A) depicts the expiratory adduction phase, while the right panel shows the inspiratory abduction phase. (B) The normal motion trajectories of bilateral vocal processes. The blue dotted lines and orange dotted lines indicate the displacement and temporal changes between the closure and opening phases for each respiratory cycle, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

performed using scatter and line plots, and temporal smoothing was applied using the Savitzky–Golay smoothing function implemented in OriginPro, resulting in smoothed motion trajectory graphs of the bilateral vocal processes (Figure 2B).

Definitions and metrics

Motion range per VF is defined as the displacement between adduction and abduction phases within a respiratory cycle. Mean motion range (MMR) is the average of motion ranges across respiratory cycles. Motion speed per VF is defined as the velocity of movement from the adduction phase to the abduction phase in a single respiratory cycle, calculated as the displacement (blue dotted lines) divided by the time interval (orange dotted lines). Mean motion speed (MMS) is the average of motion speeds across respiratory cycles.

To control for anesthetic depth and individual differences, the contralateral healthy VF in each rabbit served as the internal control. Mean motion range ratio (MMRR) = $\text{MMR}_{\text{right}}/\text{MMR}_{\text{left}}$. Mean motion speed ratio (MMSR) = $\text{MMS}_{\text{right}}/\text{MMS}_{\text{left}}$. These ratios quantify inter-individual variations in VF motion.

Transmission electron microscopy (TEM)

Tissue samples from the proximal and distal ends of the RLN injury site (approximately 1 cm in total) were collected at 2, 4, 8, and 12 weeks post-op, sectioned into 1 mm³ pieces, and fixed. Ultrathin sections (60 nm) were stained and examined by transmission electron microscope (HT7800/HT7700, HITACHI, Tokyo, Japan). Using ImageJ at 20 μm magnification, myelinated nerve fiber and axon diameters were measured, and cross-sectional areas were calculated. Myelinated axon density (MAD, the number of axons per unit area) and N-ratio (the ratio of

myelinated axon area to total nerve trunk area) were determined as indicators of nerve injury and regeneration.

Statistical analysis

SPSS 13.0 software was used for statistical analysis. The Kruskal-Wallis H test was used to compare myelinated axon densities and N-ratios across multiple independent samples. Independent samples t-tests were used to analyze outcome metrics (MMR, MMS, MMRR, and MMSR) separately. Statistical significance was set at $P < 0.05$.

RESULTS

TEM

RLN clamp and electrocoagulation injury models (Figure 3)

The clamp and electrocoagulation injury models exhibited similar trends in N-ratio and MAD changes under TEM. At 2 weeks, a large number of myelinated nerve fibers were damaged, with myelin sheaths disappearing, and a small number of regenerating myelinated fibers were observed. At 4 weeks, regenerating myelinated fibers increased gradually, though some damaged persisted. At 8 weeks, the quantity of regenerating myelinated fibers increased significantly, characterized by thin axons and thin myelin sheaths, and the MAD reached its peak. At 12 weeks, the MAD decreased, but the area of myelinated axons increased. Significant differences in N-ratio and MAD were observed in the clamp injury model between 4 and 8 weeks, and in the electrocoagulation injury model between 2 and 8 weeks (Figure 4).

RLN transection injury model (Figure 3)

At 2 weeks, the epineurium, perineurium, and endoneurial tubes were disorganized, with extensive myelin sheath loss

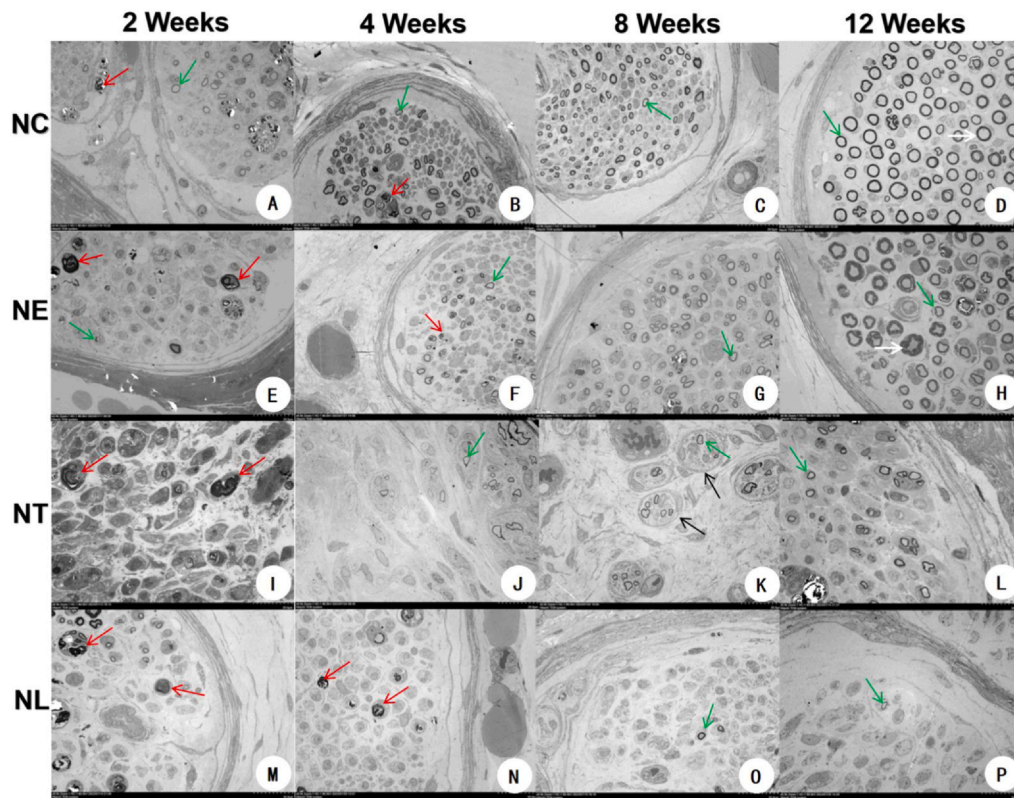


FIGURE 3. Postoperative electron microscopy of different types of RLN injury models (500 $\times$). (A–D) Clamp injury; (E–H) Electrocoagulation injury; (I–L) transection injury; (M–P) ligated injury; A,E,I,M (2 weeks postoperation); B,F,J,N (4 weeks postoperation); C,G,K,O (8 weeks postoperation); and D,H,L,P (12 weeks postoperation). Red arrows indicate disrupted myelin nodes, green arrows show regenerated thin myelin, black arrows point to regenerating perineurium, and white arrows mark thickened myelin. RLN, recurrent laryngeal nerve. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

and no regenerating myelinated nerve fibers. From 4 to 12 weeks, regenerating myelinated fibers increased, and the MAD showed a continuous increase. However, the overall number of regenerating fibers was significantly lower than in the normal control group, and the perineurium structure differed markedly from that of normal controls. Statistically significant differences in N-ratio were observed between 4 and 8 weeks (Figure 4).

RLN ligation injury model (Figure 3)

At 2 weeks, extensive damage to myelinated nerve fibers and disappearance of myelin sheaths were observed, with only a few myelinated fibers remaining. At 4 weeks, the destruction of myelin lamellae persisted, and no regenerating myelin sheaths were observed. From 8 to 12 weeks, only sporadic regenerating myelinated fibers were present. The MAD and N-ratio exhibited a continuous decline from 2 to 12 weeks (Figure 4).

Comparison of N-ratio and MAD among four models (Figure 4)

Statistically significant differences were observed between clamp and transection injuries at all postoperative time

points. For clamp and ligation injuries, significant differences were found at all time points except at 2 weeks. Clamp and electrocoagulation injuries showed statistically significant differences only at 2 weeks. Comparisons between electrocoagulation and ligation injuries revealed statistically significant differences at 8 and 12 weeks. No statistically significant differences were observed between electrocoagulation and transection injuries or between transection and ligation injuries at any time point.

VF motion assessment results

At 2 weeks, no VF motion recovery was observed in any of the models. In the clamp injury model, VF motion recovery began at 4 weeks. In the electrocoagulation injury model, all cases except one showed recovery beginning at 4 weeks. In the transection injury model, no recovery was observed except for one case that demonstrated synkinesis at 12 weeks. Synkinesis was identified through laryngoscopic observation, the affected VF adducted during inhalation while abducted during exhalation. This paradoxical movement indicates misdirected regeneration of the nerve. In the ligation injury model, no recovery of VF motion or synkinesis was observed even at 12 weeks.

Only the nerve clamp and nerve electrocoagulation injury models showed recovery of VF movement, so our VF

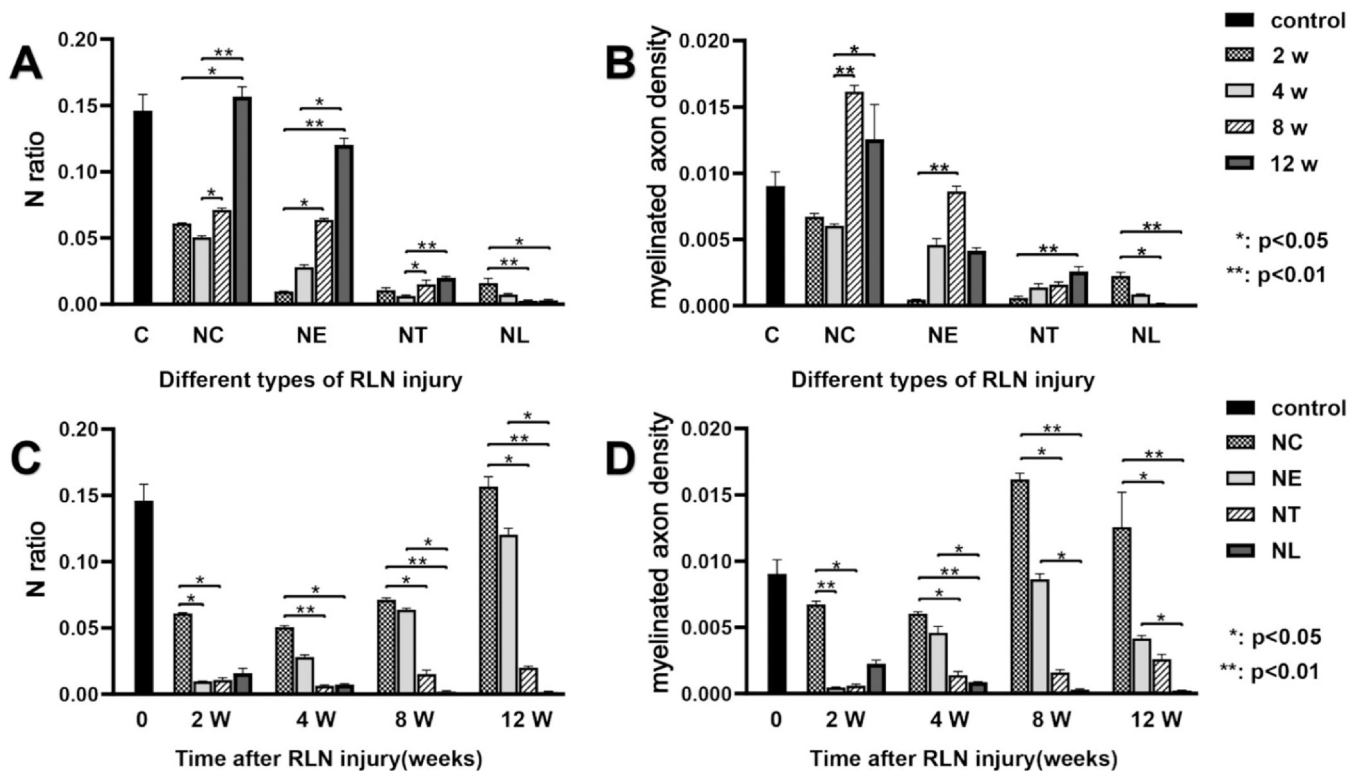


FIGURE 4. Comparison of N-ratio and MAD among different types of RLN injury models. (A, B) A comparison of the differences in N ratio and myelinated axon densities observed across four distinct types of RLN injury models, evaluated at different postoperative time points. (C, D) Comparisons of differences in myelinated axon densities and N ratios at the same time point for four different types of RLN injury models. MAD, myelinated axon density; RLN, recurrent laryngeal nerve.

movement assessment was limited to these two models. Preoperatively, bilateral VF motion was symmetrical, with identical motion ranges and speeds (Figure 5A). At 2 weeks, no motion recovery was observed (Figure 5B). At 4 weeks, motion recovery was observed. However, the range was approximately 22% smaller ($t = 9.369$, $P < 0.01$) and the speed was about 58% slower ($t = 4.961$, $P < 0.05$) than that of the normal side (Figure 5C). At 8 weeks, the range was approximately 7% smaller ($t = 5.657$, $P < 0.05$) and the speed was about 41% slower ($t = 4.507$, $P < 0.05$) than that of the normal side (Figure 5D), the differences between the two sides had narrowed. At 12 weeks, the motion range difference had further narrowed to about 2%, and this difference was no longer statistically significant ($t = 1.225$, $P > 0.05$). However, the speed difference persisted, narrowing to approximately 24% and remaining statistically significant ($t = 8.820$, $P < 0.01$). These findings suggest that recovery of motion range occurs earlier than recovery of velocity.

Statistically significant differences in MMRR (Figure 5E) and MMSR (Figure 5F) were observed between the preoperative period and 2 weeks, as well as between 2 and 4 weeks. Significant differences in MMRR were observed between 4 and 8 weeks, while no statistically significant differences in MMRR or MMSR were observed between 8 and 12 weeks.

In one case of electrocoagulation injury at 8 weeks and one case of transection injury at 12 weeks, the affected side exhibited synkinesis.

Simultaneous electron microscopy of the RLN revealed abundant regenerated myelinated nerve fibers distal to the injury site. However, significant differences in the structure of nerve fascicles were observed between the proximal and distal ends of the injury site (Figure 6).

DISCUSSION

Different degrees of nerve damage present varying challenges to regeneration and necessitate tailored repair strategies. However, previous research on RLN damage and regeneration mechanisms from various injuries is limited.¹⁷⁻¹⁹ The mechanisms of VF motion recovery are unclear, hindering targeted treatment development and negatively affecting patient prognosis.

Upon examining the recovery of VF motion, this study revealed that the most significant differences were between clamp and ligation injuries. In the clamp injury model, VF motion initiated its recovery process by 4 weeks and achieved nearly complete restoration by 8 weeks. No VF motion recovery or synkinesis were observed in the ligation injury model by 12 weeks. Electron microscopy results showed that a small number of regenerating myelinated

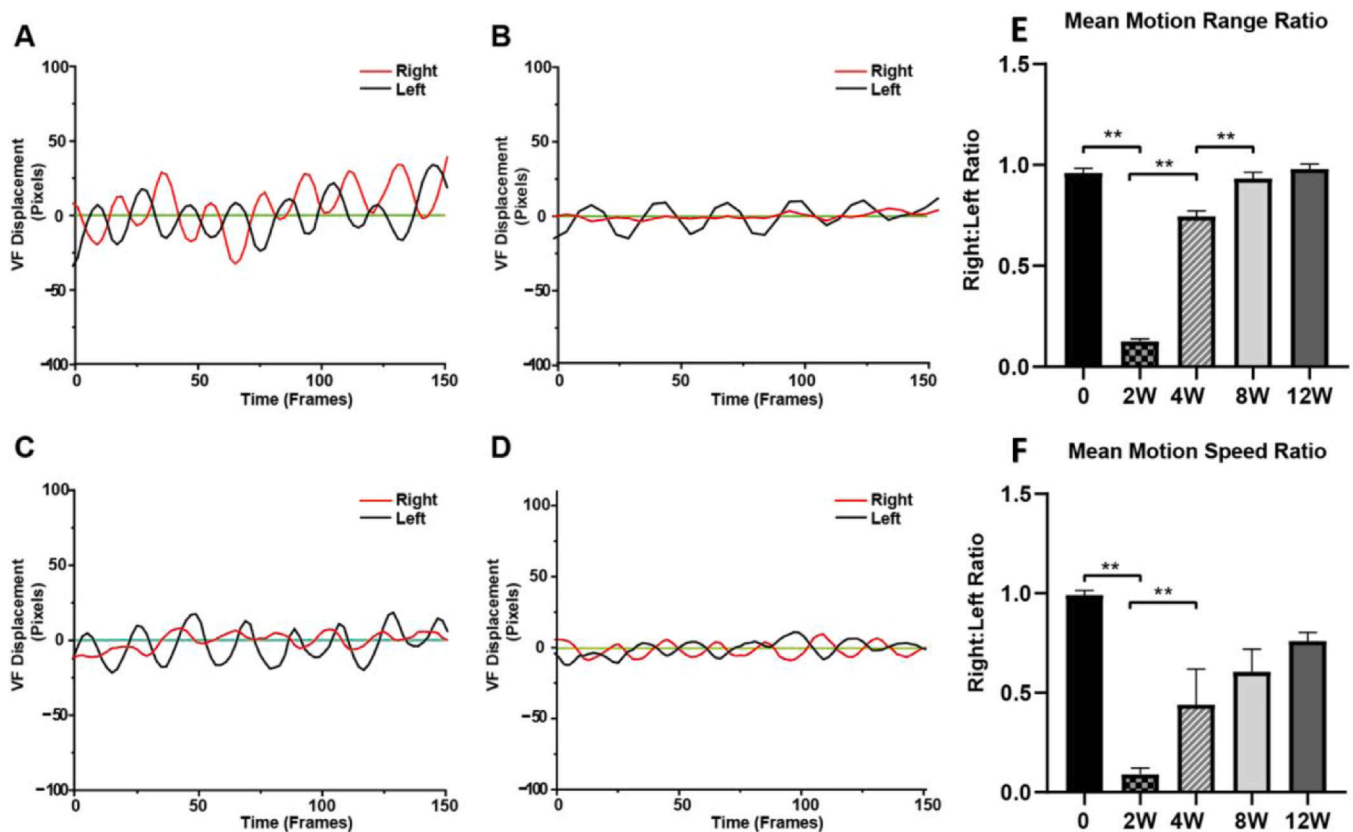


FIGURE 5. Time-dependent changes in motion range and speed of the bilateral VFs in recovery models of NC and NE. (A–D) The motion trajectories of the bilateral VFs at different time points in nerve clamp injury models: preoperation (A), 2 weeks postoperation (B), 4 weeks post-operation (C), and 8 weeks postoperation (D). The black line represents the left VF, and the right VF by the red line. The x-axis represents the video frame number, and the y-axis represents VF displacement, measured in pixels. (A) Preoperative motion during inhalation and exhalation, showing symmetrical oscillatory movement of the bilateral VFs along their respective midlines (green line). (B) At 2 weeks postoperation, the affected side remains fixed compared to the normal side. (C) At 4 weeks postoperation, the motion range of the affected side is smaller, and its speed is slower than that of the normal side. (D) At 8 weeks postoperation, the motion range and speed of the affected side increase, becoming closer to the normal side. (E, F) Differences in MMRR and MMSR at various time points. ** denotes $P < 0.01$. MMRR, mean motion range ratio; MMSR, Mean motion speed ratio; NC, nerve clamp injury model; NE, nerve electrocoagulation injury model; VF, vocal fold. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

axons appeared at 2 weeks after clamp injury, with a gradual increase in regenerating myelinated axons by 4 weeks and reaching the maximum density of myelinated axons at 8 weeks. Conversely, ligation injury exhibited continuous disruption of myelin lamellae at 4 weeks, with no evidence of nerve regeneration, and only sporadic regenerating myelinated axons were observed at 12 weeks. At all time points except 2 weeks, the N-ratio and MAD were significantly lower in the ligation injury model than in the clamp injury model. These results suggest that clamp injuries are mild, mostly reversible and allow for rapid nerve regeneration, which conforms to the type II of Sunderland classification and is consistent with previous studies.^{20–23} In contrast, ligation injuries are severe, with low regeneration levels, which conforms to type IV of the Sunderland classification. Low regeneration levels are possibly due to regenerating axons being unable to cross the ligature and enter the distal endoneurial tubes. The superior prognosis

of clamp injuries over ligation injuries aligns with the results of Mu L.¹⁸ These findings offer significant clinical guidance: clamp injuries can be treated conservatively through observation, neurotrophic therapy, and rehabilitation, whereas ligation injuries necessitate prompt intervention, including suture removal and nerve decompression.

Electrocoagulation injuries demonstrated significant interindividual variability in VF recovery, which may depend on the proximity of thermal damage to the nerve. We standardized the injury site, power, and duration to minimize variability in injury severity. However, differences in recovery outcomes may be due to the difficulty in precisely controlling the depth and extent of nerve injury. Compared to ligation injuries, electrocoagulation injuries had higher N-ratio and MAD at 4, 8, and 12 weeks, with statistically significant differences in MAD ($P < 0.05$). Still, these differences were less pronounced than those observed between

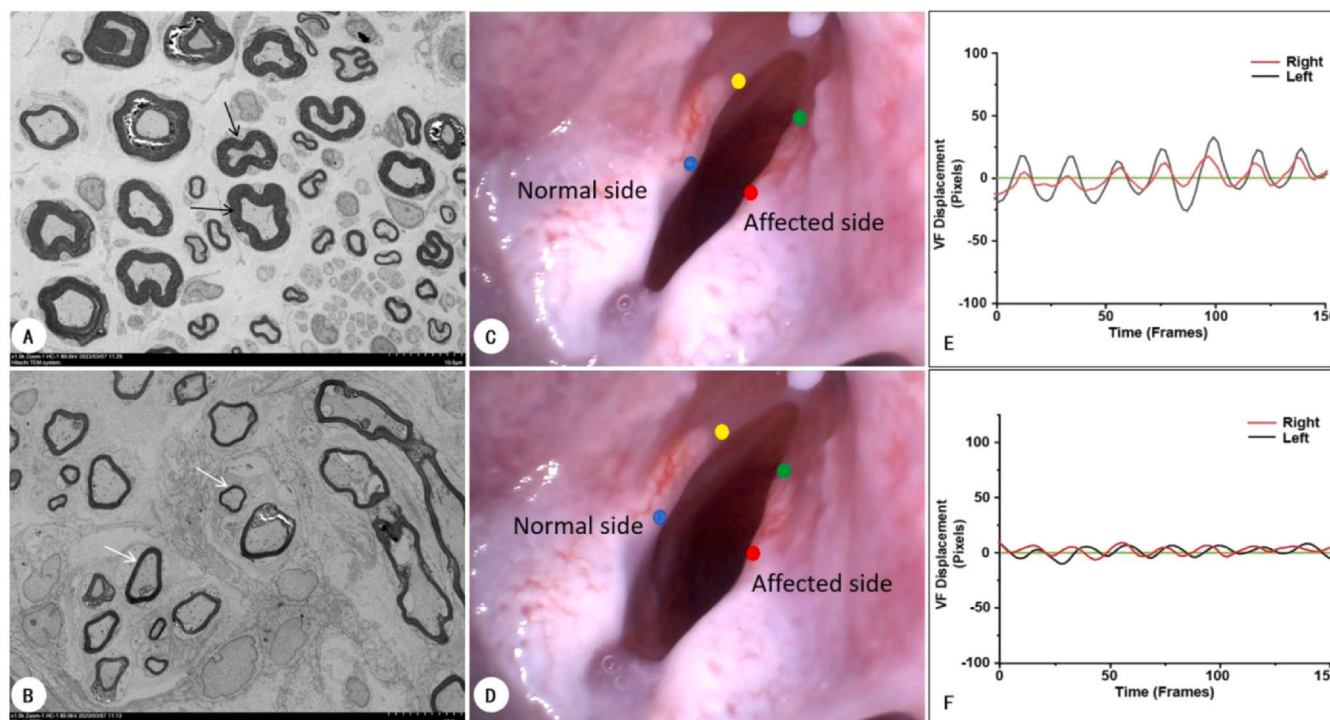


FIGURE 6. Electron microscopy, laryngoscopic images, and motion trajectory graphs of the RLN and VFs in the transection injury model with synkinesis at 12 weeks postoperation. (A, B) Electron microscopic images of the proximal and distal ends, respectively, at the injury site (1000 $\times$). The black arrows denote intact myelin sheaths, whereas the white arrows indicate regenerated myelin sheaths, which are thinner than usual. (A, B) Significant differences in the structure of nerve fascicles between the proximal and distal ends. (C, D) Laryngoscopic images of VF motion during minimal closure in exhalation (C) and maximal opening in inhalation (D). Blue dots mark the normal side vocal process; red dots mark the affected side vocal process; yellow and green dots indicate the midpoints of the normal and affected VFs, respectively. (E, F) Motion trajectory graphs of the bilateral VF midpoints (E) and vocal processes (F). In F, asynchronous motion of the vocal processes is evident, while E shows synchronous motion of the VF midpoints. The motion range of the affected side is smaller than that of the normal side, indicating the presence of synkinesis. RLN, recurrent laryngeal nerve; VF, vocal fold. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

clamp and ligation injuries. Clamp and electrocoagulation injuries differed significantly only at 2 weeks in the N-ratio and MAD. These findings suggest that electrocoagulation injuries are more severe than clamp injuries but less severe than ligation injuries, and the damage degree conforms to type VI (a mixed injury pattern of types II-IV), which was proposed by Mackinnon and Dellon²⁴ based on the Sunderland classification. Additionally, the significant inter-individual variability in electrocoagulation injuries complicates prognosis, emphasizing the need for comprehensive assessments that consider injury location, depth, and electromyographic findings to guide clinical management.

Further evaluating models of VF recovery in clamp and electrocoagulation injuries showed that the affected VF exhibited a progressive increase in motion range and speed from 4 weeks to 8 weeks, and the differences between the folds narrowed. By 12 weeks, the difference in motion range had decreased to approximately 2%, while the difference in motion speed remained at approximately 24%. These results indicate that the recovery of motion range precedes that of speed, with asynchronous motion being a

characteristic feature of RLN injury repair. Simultaneous TEM showed an increasing trend in the number of regenerating myelinated axons and N-ratio between 4 and 8 weeks, peaking at 8 weeks. This corresponded to the observed improvement in VF motion range and speed. Between 8 and 12 weeks, the number of regenerating myelinated axons and the N-ratio began to decline, coinciding with reduced changes in VF motion. This phenomenon may be related to the degeneration of early mismatched axons.¹⁹ The dynamic changes in VF motion amplitude and speed, as observed via stroboscopic laryngoscopy, may serve as indicators for predicting nerve injury prognosis and guiding treatment.

For transection injuries, poor VF motion recovery corresponded to severe nerve damage and limited nerve regeneration. Only one case demonstrated synkinesis at 12 weeks. Electron microscopy revealed significant differences in nerve bundle structure between proximal and distal sites. Regenerating axons unable to reenter the original endoneurial tubes, which may result in synkinesis. Nishimoto *et al.*¹² also found synkinesis in a murine model of RLN transection and suggested that the small number of

regenerating nerve fibers and misdirected regeneration may be the main reason for the poor recovery of VF movement. Clamp and ligation injuries did not exhibit synkinesis. This may be related to the mild injury and complete recovery of clamp injuries and the limited regeneration in severe ligation injuries. Similarly, van Lith-Bijl JT¹⁹ reported good recovery of clamp injuries, whereas transection injuries failed to restore significant function despite nerve repair. These findings underscore the severity of transection injuries, which conforms to type V of the Sunderland classification. Slow regeneration and predisposition to synkinesis highlighting the need for strategies to promote regeneration and prevent mismatched axon growth.

CONCLUSION

The severity of RLN injuries decreases in the following order: ligation > transection > electrocoagulation > clamp. Recovery of VF motion range precedes that of motion speed, and asynchronous motion is a characteristic feature of RLN injury repair.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jvoice.2026.01.014](https://doi.org/10.1016/j.jvoice.2026.01.014).

References

1. Foerster Gerhard, Mueller, et al. Laryngeal EMG: Preferential damage of the posterior cricoarytenoid muscle branches especially in iatrogenic recurrent laryngeal nerve lesions. *Laryngoscope*. 2018;128:1152–1156.
2. Chen CC, Huang YC, Lee ST, et al. Long-term result of vocal cord paralysis after anterior cervical disectomy. *Eur Spine J*. 2014;23:622–626.
3. Ta JH, Liu YF, Krishna P. Medicolegal aspects of iatrogenic dysphonia and recurrent laryngeal nerve injury. *Otolaryngol Head Neck Surg*. 2016;154:80–86.
4. Hernandez-Morato I, Valderrama-Canales FJ, Berdugo G, et al. Reorganization of laryngeal motoneurons after crush injury in the recurrent laryngeal nerve of the rat. *J Anat*. 2013;222:451–461.
5. Wang B, Yuan J, Chen X, et al. Functional regeneration of the transected recurrent laryngeal nerve using a collagen scaffold loaded with laminin and laminin-binding BDNF and GDNF. *Scientific Rep*. 2016;6:32292.
6. Tessema B, Pitman MJ, Roark RM, et al. Evaluation of functional recovery of recurrent laryngeal nerve using transoral laryngeal bipolar electromyography: a rat model. *Ann Otol Rhinol Laryngol*. 2008;117:604–608.
7. Tessema B, Roark RM, Pitman MJ, et al. Observations of recurrent laryngeal nerve injury and recovery using a rat model. *Laryngoscope*. 2009;119:1644–1651.
8. Monaco GN, Brown TJ, Burgette RC, et al. Electrical stimulation and testosterone enhance recovery from recurrent laryngeal nerve crush. *Restor Neurol Neurosci*. 2015;33:571–578.
9. Hernandez-Morato I, Sharma S, Pitman MJ. Changes in neurotrophic factors of adult rat laryngeal muscles during nerve regeneration. *Neuroscience*. 2016;333:44–53.
10. Hernandez-Morato I, Valderrama-Canales FJ, Berdugo G, et al. Reorganization of laryngeal motoneurons after crush injury in the recurrent laryngeal nerve of the rat. *J Anat*. 2013;222:451–461.
11. Choi JS, Oh SH, An HY, et al. Functional regeneration of recurrent laryngeal nerve injury during thyroid surgery using an asymmetrically porous nerve guide conduit in an animal model. *Thyroid*. 2014;24:52–59.
12. Nishimoto K, Kumai Y, Yumoto E. Paradoxical movement of rat vocal folds following recurrent laryngeal nerve injury. *Acta Otolaryngologica*. 2014;134:1164–1171.
13. Pennington-FitzGerald W, Joshi A, Honzel E, et al. Development and application of automated vocal fold tracking software in a rat surgical model. *Laryngoscope*. 2024 Jan;134:340–346.
14. Haney MM, Hamad A, Leary E, et al. Automated quantification of vocal fold motion in a recurrent laryngeal nerve injury mouse model. *Laryngoscope*. 2019;129:E247–E254.
15. Natick, Massachusetts, USA: The MathWorks, Inc.; 2022.
16. Northampton, Massachusetts, USA: OriginLab Corporation.; 2021.
17. Xu Wen, Han Demin, Hu Huiying, Fan Erzhong. Characteristics of experimental recurrent laryngeal nerve surgical injury in dogs. *Ann Otol Rhinol Laryngol*. 2009;118:575.
18. Mu L, Yang S. An experimental study on the laryngeal electromyography and visual observations in varying types of surgical injuries to the unilateral recurrent laryngeal nerve in the neck. *Laryngoscope*. 2010;120(7 Pt 1):699–708.
19. Van Lith-Bijl JT, Mahieu HF, Stolk RJ, et al. Laryngeal abductor function after recurrent laryngeal nerve injury in cats. *Arch Otolaryngol Head Neck Surg*. 1996;122:393.
20. Fung K, Hogikyan ND, Heavner SB, et al. Development and characterisation of an experimental recurrent laryngeal nerve injury model for the study of viral gene therapy. *J Laryngol Otol*. 2008;122:500–505.
21. Paniello RC, Rich JT, Debnath NL. Laryngeal adductor function in experimental models of recurrent laryngeal nerve injury. *Laryngoscope*. 2015;125:E67–E72.
22. Müns S. Prevention of recurrent laryngeal nerve paralysis by demonstration of the nerve during thyroid surgery. *Laryngo-Rhino-Otologie*. 2005;84:261–265.
23. Belachew Tessema, Pitman Michael J, Roark Rick M, et al. Evaluation of functional recovery of recurrent laryngeal nerve using transoral laryngeal bipolar electromyography: a rat model. *Ann Otol Rhinol Laryngol*. 2008;117:604–608.
24. Mackinnon SE, Dellon AL. Nerve repair and nerve grafts. In: Mackinnon SE, ed. *Surgery of the Peripheral Nerve*. New York, NY, USA: Thieme; 1988.